

Al-Ti Alloys:

Main properties of main element:

Aluminum has low density, giving it a high strength to weight ratio, making parts relatively light. Some other properties include: high corrosion resistance and stability at high temperatures making it useful for industries like aerospace.

Main strengthening methods:

Precipitation hardening by heat treatment: This forms precipitates in the aluminum crystals that are harder for dislocations to move

Work hardening: Cold working the metal which "tangles up" the dislocations locking them into shape and making it harder for them to move.

Role played by main alloying element:

Ti atoms dissolve into the aluminum, strengthening the crystal lattice by slowing the dislocation slip, therefore, increasing yield strength. The Ti atoms also shrink grain sizes making the metal tougher since the grain boundaries make it harder for dislocations to move through the metal.

Main engineering applications and properties:

These alloys are very useful in the aerospace and automotive sectors. Structures such as aircraft fuselage require high strength to weight ratio to save fuel and high corrosion resistance due to the extreme environments they operate in.

Individual sustainability factors:

The production of aluminum is energy intensive (2-3x more than typical steel resulting in higher CO₂ emissions. However due to the fact that aluminum parts are light weight, they help with massive fuel savings in airplanes and cars. Similarly, titanium has high extraction and processing costs and emissions. Similarly, both require a lot of water during mining and processing. Aluminum, however, has high recycling rates (around 80% in the transport sector), recycling is much less energy intensive than production. Titanium is also easily recycled, however its recycling rate is not as high as that of aluminum.

Ti-Al Alloys:

Main properties of main element:

Titanium has a low density, resulting in a very high strength to weight ratio (even higher than that of aluminum), high corrosion resistance because of its oxide layer and high stability. An interesting fact about titanium is that its biocompatibility making it one of the best options to use as implants in the human body.

Main strengthening methods:

Solid solution strengthening - dissolving Al atoms into the structure to increase strength.

Role played by main alloying element:

Al atoms dissolve into the titanium crystal lattice, distorting it, and in turn making dislocation slip harder, increasing yield strength significantly.

Main engineering applications and properties:

Ti-Al Alloys are used in high performance sectors. An example is jet engine compressor blades. This is because these alloys have a high strength to weight ratio (this not only makes the blades wider but also reduces the centrifugal force they create), high fatigue resistant and creep stability at the high temperatures that compressor blades experience.